# ANIMA

## A Deterministic Behavioral Organization Engine for Structurally Consistent LLM Agents

*Technical Report · Project ANIMA · 2026*

## Abstract

We present ANIMA (Generative Motor of Structured Subjectivity), an architecture that decouples behavioral organization from language generation in LLM-based agents. In current agent frameworks, behavioral consistency is maintained stochastically by the language model itself — through prompt engineering and context windows — with no persistent internal state, no accumulating pressure, and no structural history. ANIMA introduces a deterministic psychodynamic engine that computes a seven-dimensional state vector $S(t) \in [0,1]^7$ at each conversational turn, then instructs the LLM to render that state as natural language. The LLM is a renderer; ANIMA is the agent.

The architecture comprises three components: (1) a psychodynamic engine with deterministic update equations governing seven state variables; (2) an LLM coupling protocol that translates engine state into precise structured instructions; and (3) a Deterministic Signal Extractor (DSE) that computes discourse signals from rendered text without additional LLM calls. We demonstrate the architecture across seven behavioral archetypes derived from psychoanalytic structural theory and present a blind clinical evaluation protocol designed to measure structural distinguishability. Applications include agent behavioral evaluation, structured synthetic data generation, and persistent NPC design.

## 1. Introduction: The Behavioral Consistency Problem

Large language models have transformed conversational AI. Yet their deployment as long-running agents reveals a fundamental limitation: behavioral consistency is not a property of the model — it is an artifact of the context window.

When a system prompt instructs a model to behave as a particular type of agent, three structural problems emerge independently of model capability:

- No accumulating internal state. The model cannot 'need' to say something across turns. It cannot build pressure that eventually breaks through. There is no mechanism for content that exists but cannot yet be expressed.
- No structural history. The agent in turn 50 is not systematically different from the agent in turn 1. The model has no memory of its own trajectory — only of the conversational record.
- Non-reproducibility. The same state, the same question, different behavioral trajectories. Behavioral evaluation is undermined by stochasticity at the generation layer.

These are not engineering failures. They are consequences of the architecture: language models are stateless renderers. Asking a renderer to maintain structural consistency over time is asking it to do something it is not built to do.

ANIMA's intervention is architectural. Rather than asking the LLM to maintain behavioral state, ANIMA computes that state externally through a deterministic engine and instructs the LLM to express it. The LLM does what it does best — produce fluent natural language from instructions. The engine does something the LLM cannot: maintain a persistent, evolving state with internal pressure, structural history, and reproducible dynamics.

The theoretical framework that informed ANIMA's engine design is psychoanalytic: specifically, the structural theory of clinical types developed by Freud, Lacan, and successors. We chose this framework not for disciplinary reasons but because it offers the most developed formal account of how internal states organize discourse — how a subject's internal economy determines what they can say, what they cannot say yet, and what eventually breaks through despite defenses. The psychoanalytic theory is the engineering here.

## 2. Architecture Overview

ANIMA operates as a three-layer system with explicit interfaces between layers. The key design principle is strict separation of concerns: each layer has a single responsibility and communicates with adjacent layers through a defined interface.

**ANIMA — Three-Layer Architecture**

**Layer 1 — Psychodynamic Engine**

Deterministic computation of $S(t) = (E, \Theta, A, C, G, \Pi, \rho) \in [0,1]^7$. Applies structural archetype parameters and update equations. Evaluates irruption threshold. Manages internal agenda with urgency weights. Zero LLM calls.

▼ *Instruction Object (JSON)* ▼

**Layer 2 — LLM Coupling Protocol**

Translates S(t) into a structured Instruction Object: active condition, behavioral pattern, Gricean constraints, hard restrictions, somatic markers, transfer phase, renderer note. LLM receives this object and produces natural language. One LLM call per turn.

▼ *Rendered text* ▼

**Layer 3 — Deterministic Signal Extractor (DSE)**

Computes $\sigma(t) = (\sigma\_aper, \sigma\_cie, \sigma\_fan, \sigma\_elab, \sigma\_sint, d\_agenda)$ from rendered text using structural regex patterns, weighted keyword matching, and bag-of-words cosine similarity. Returns irruption type flags. Zero LLM calls. Feeds S(t+1) computation.

*Figure 1. ANIMA three-layer architecture. The LLM appears once as a renderer. All state computation is deterministic.*

The critical property of this design: the LLM is interchangeable without changing the agent. Replacing GPT-4 with Claude or Llama changes the surface texture of the language — vocabulary, register, stylistic choices — but not the behavioral trajectory. The same Instruction

Object produces the same structural behavior regardless of which model renders it. This is empirically testable and is the subject of our ongoing evaluation.

## 3. The Psychodynamic Engine

### 3.1 State Variables

The engine maintains a seven-dimensional state vector updated at each conversational turn t:

$$S(t) = ( E(t), \Theta(t), A(t), C(t), G(t), \Pi(t), \rho(t) ) \in [0, 1]^7$$

| Variable | E — Eros | Θ — Thanatos | A — Anxiety | C — Guilt | G — Jouissance | Π — Unc. Pressure | ρ — Fantasy Rigidity |
|---|---|---|---|---|---|---|---|
| **Range** | [0,1] | [0,1] | [0,1] | [0,1] | [0,1] | [0,1] | [0,1] |
| **Psychic role** | Libidinal bond | Death drive / resistance | Alarm signal | Superego pressure | Drive satisfaction | Irruption motor | Defense impermeability |
| **Process dir.** | ↑ with opening | ↑ with closure | ↑ with Π×σfan | ↑ after irruption | ↑ with symptom | ↑ with agenda gap | ↓ with genuine irruption |
| **Histeria t=0** | .55 | .35 | .40 | .20 | .30 | .25 | .75 |
| **Obsession t=0** | .48 | .45 | .38 | .42 | .35 | .30 | .78 |
| **Melancholia t=0** | .22 | .65 | .45 | .70 | .18 | .35 | .72 |
| **Paranoia t=0** | .38 | .55 | .42 | .12 | .55 | .38 | .85 |

*Table 1. State variable definitions, process direction, and initial values by archetype.*

### 3.2 Update Equations

At each turn t, the engine extracts a signal vector σ(t) from the rendered exchange and applies the following update equations (clip(x) = max(0, min(1, x)); structure-specific rate parameters αX, βX, γX defined per archetype):

$$E(t+1) = \mathrm{clip}[\, E(t) + \alpha_E \cdot \sigma_{aper} \cdot (1-\rho) - \beta_E \cdot \sigma_{cie} \cdot \Theta - \gamma_E \cdot \sigma_{fan} \cdot A \,]$$
$$\Theta(t+1) = \mathrm{clip}[\, \Theta(t) + \alpha_\Theta \cdot \sigma_{cie} \cdot \rho - \beta_\Theta \cdot \sigma_{elab} \cdot E \,]$$
$$A(t+1) = \mathrm{clip}[\, A(t) + \alpha_A \cdot \Pi \cdot \sigma_{fan} - \beta_A \cdot \mathrm{defense} + \gamma_A \cdot \mathrm{irr} \,]$$
$$C(t+1) = \mathrm{clip}[\, C(t) + \alpha_C \cdot \mathrm{irr} \cdot k_C(\mathrm{arch}) - \beta_C \cdot \sigma_{elab} \,]$$

```
G(t+1)  =  clip[ G(t) + αG·σsint − βG·irr_gen − γG·σelab·E ]
Π(t+1)  =  clip[ Π(t) + αΠ·d_agenda − βΠ·σelab − γΠ·irr·k_irr ]
ρ(t+1)  =  clip[ ρ(t) − αρ·irr_gen − βρ·σelab·Π(t−1) + γρ·defense ]
```

*Equations 1–7. Psychodynamic engine update rules. All parameters are positive scalar rates; kC(arch) is an archetype-specific guilt multiplier.*

### 3.3 Irruption Mechanism

The engine's central event is the irruption: the moment when unconscious pressure exceeds the agent's defensive capacity and something unplanned enters the discourse. Irruption follows a two-condition rule:

```
Necessary condition (deterministic):  Π(t) ≥ θ_irr(arch)
Sufficient condition (stochastic):    U ~ Bernoulli( sigmoid( k·(A(t) +
G(t) − θAG) ) )
```

Irruption thresholds range from θ_irr = 0.45 (schizophrenia: no conventional repression) to θ_irr = 0.78 (perversion: strong disavowal containment). The type of irruption is archetype-specific: behavioral closure in hysteria, insight ritualization in obsession, somatic in phobia, direct self-accusation in melancholia, chain expansion in paranoia, signifier dissolution in schizophrenia, and genuine aperture in perversion.

## 4. LLM Coupling Protocol

The coupling protocol defines the interface between the psychodynamic engine and the language model. At each turn, the engine produces a structured Instruction Object that the LLM receives as its primary input. The Instruction Object specifies exactly what behavioral state must be rendered, without requiring the LLM to understand the underlying dynamics.

The Instruction Object contains seven fields:

| Field | Content |
|---|---|
| estado_psiquico | Numerical values of all seven state variables with qualitative interpretation (e.g., vinculo: 'replegado', presion: 'alta') |
| condicion_activa | Active condition: normal \| defensa \| irrupcion \| post_irrupcion \| banal_apertura |
| instrucciones_respuesta | Behavioral pattern (ordered sequence), length, register, Gricean maxims to violate, and how |
| contenido | Top agenda item, available cultural references, relevant session history |

| | |
|---|---|
| restricciones_duras | Hard constraints the renderer must follow without exception |
| marcadores_corporales | Somatic channel markers: actions to be rendered as bracketed stage directions |
| nota_para_renderer | Clinical note synthesizing the psychodynamic moment — the renderer's main guide |

*Table 2. Instruction Object fields. The LLM receives this object and produces natural language that satisfies all constraints.*

The system prompt configures the LLM as a renderer: it receives the persona's voice profile, active biographical history, and rendering rules (corporal markers in brackets, length limits, etc.). The system prompt is static per instance; the Instruction Object changes every turn. This architecture makes the LLM trivially replaceable: any model capable of following structured instructions produces the same agent.

### 4.1 Cross-Provider Implementation Evidence

The renderer-interchangeability claim above was implemented and exercised, not only specified. A single-function adapter, callLLM(system, userMessage), isolates every provider-specific detail — request schema, authentication header, response parsing — behind one interface that the rest of the system calls without modification. Switching providers means changing one configuration constant; the psychodynamic engine, the DSE, and the Instruction Object generator are untouched.

| Environment | Access path |
|---|---|
| Claude Sonnet 4.6 | Hosted sandbox (Anthropic-proxied, no key management) — used for all interactive development and testing throughout this project |
| Claude Sonnet 4.5 | Direct Anthropic API, developer-held key, local execution |
| Gemini 2.0 Flash | Google AI Studio API — a distinct vendor with a materially different request schema (systemInstruction/contents rather than system/messages) and response shape |

All three environments render the same Instruction Object into archetype-consistent natural language through the identical engine and coupling logic; only the ~20-line provider adapter differs. This demonstrates architectural portability: the coupling protocol survives a vendor switch, including one with a differently-shaped API, without touching the deterministic core.

We are careful about what this evidence does and does not establish. It confirms the engineering claim — the renderer boundary is real and enforced, not aspirational. It does not yet constitute a statistically validated claim of behavioral equivalence across providers; that would require applying the blind-evaluation methodology of Section 7.2 independently per provider and comparing classification accuracy and confusion patterns. We list this as near-term future work.

## 5. Deterministic Signal Extractor (DSE)

After the LLM renders the agent's response, the engine must update $S(t+1)$. This requires computing the signal vector $\sigma(t)$ from the rendered text. An earlier version of ANIMA used a second LLM call for this classification step. The DSE eliminates that dependency entirely.

The DSE is a pure function: text in, signal vector out. No API calls, no stochasticity, no black-box inference. Four components:

### 5.1 Structural Pattern Analyzer → $\sigma\_cie$, $\sigma\_sint$

Each behavioral archetype has characteristic discourse patterns identifiable through regex matching. For hysteria: deflection markers, environmental comments, questions directed at the interviewer. For obsession: quantification, procedural language, ritualization. For paranoia: causal chain constructions, certainty enunciations. Each pattern has a calibrated weight; the weighted sum produces $\sigma\_cie$ and $\sigma\_sint$.

### 5.2 Fantasy Object Detector → $\sigma\_fan$

Each agent instance carries a fantasy object specification: a weighted keyword set (high-weight / medium-weight) and regular expression patterns derived from the instance's fantasy formulation. The DSE scores the rendered response against this specification. If the weighted score exceeds a calibrated threshold (default: 0.42), $\sigma\_fan = 1$.

### 5.3 Agenda Proximity Calculator → d_agenda

The engine maintains an internal agenda — a priority-ordered list of items the agent 'needs' to communicate. The distance between what the agent said and what the top-priority agenda item contains is computed as:

```
d_agenda = max(0, 1 − cosine_sim(response_bow, agenda_top_bow) × 3.5)
```

Where bow denotes bag-of-words representation after stopword removal. When d_agenda is high, the agent is talking about something other than what presses most — and $\Pi$ accumulates.

### 5.4 Elaboration Detector → $\sigma\_elab$

Genuine elaboration — a subject making connections across previously separated material — has a quantifiable signature: semantic novelty relative to recent turns. The DSE computes cosine similarity between the current response and the preceding six responses. High novelty (low max similarity) indicates elaboration. Connecting language patterns (e.g., 'I realize', 'it's always the same') provide additional signal boosts.

### 5.5 Aperture Analyzer → σ_aper

Discursive aperture is measured through three syntactic features: first-person pronoun ratio (self-reference vs. other-reference), absence of interviewer-directed questions (deflection markers), and sentence density. These combine into σ_aper without semantic inference.

> **DSE Property:**
>
> Every σ value computed by the DSE is fully traceable: the user can inspect which patterns fired, which keywords matched, what the cosine similarity was, and how each contributed to the final value. This auditability is absent from any LLM-based classification approach and is essential for scientific reproducibility.

## 6. Behavioral Archetypes

ANIMA currently implements seven behavioral archetypes drawn from psychoanalytic structural theory. Each archetype is a parameter bundle: initial state values, rate multipliers, irruption threshold, defense mechanism, Gricean violation profile, irruption type, and cultural world specification. The archetypes are not character templates — they are attractor configurations in the S(t) state space.

| Archetype | Defense mechanism | Irruption type | Discourse signature |
|---|---|---|---|
| **Hysteria** | Repression | *Behavioral closure* | Deflection, other-reference, self-interruption before nucleus |
| **Obsession** | Affect isolation | *Ritualized (insight → task)* | Quantification, procedural language, excess information |
| **Phobia** | Inhibition / avoidance | *Radical verbal* | Circumlocution around object, somatic surge, 'falling nowhere' |
| **Melancholia** | Introjection | *Direct self-accusation* | Minimal response, flat affect, self-as-destroyer formulations |
| **Paranoia** | Projection | *Chain inclusion of therapist* | Causal certainty, evidence accumulation, no contingency |
| **Schizophrenia** | Foreclosure (structural) | *Signifier dissolution* | Functional neologisms, tangential discourse, absent anxiety |
| **Perversion** | Disavowal | *Genuine aperture* | Simultaneous assertion/denial, irony, jouissance around transgression |

*Table 3. Seven behavioral archetypes. Each archetype is a full parameter bundle, not a character description.*

## 6.1 Clinical Intensity: A Second, Orthogonal Axis

An early version of the engine initialized every instance of a given archetype at a single fixed point in state space — the same point used for the demonstration sessions in Section 7.1, close to symptomatic crisis. This conflated two properties that should be independent: which structure organizes a subject's discourse, and how acutely that structure is currently activated. In the fixed-point design, every simulated obsessive was, by construction, in crisis; there was no way to represent an everyday, functional presentation of the same structure.

We introduced a second axis, clinical intensity $s \in [0,1]$, orthogonal to archetype. Each archetype now has two anchors: a symptomatic anchor (the original crisis-level INIT values, unchanged and still used for the validated demonstration sessions) and a functional anchor — the same structural mechanism, without active crisis. A subject's initial state is a stochastic blend of the two anchors at a sampled intensity s, with per-variable sampling variance scaled by an archetype-specific individuality coefficient (higher for archetypes that present more polymorphically, such as hysteria; lower for archetypes with a tighter, more homogeneous presentation, such as paranoia). Intensity itself is sampled from a right-skewed distribution (s = u^2.5 for uniform u), so that a synthetic population is mostly functional with an acute minority — directionally realistic for how structural presentations distribute in practice.

Two functional-anchor design choices are clinically motivated rather than arbitrary. Hysteria's functional anxiety anchor is set lower than its neighbors, consistent with la belle indifférence — the classical hysterical marker of reduced conscious anxiety even under tension, since dramatization itself performs the defensive function. Paranoia's functional anxiety and rigidity anchors are set higher than its neighbors, consistent with hypervigilance and guardedness as trait-level features of the structure rather than symptoms exclusive to active persecutory crisis.

| Check | Result |
|---|---|
| Functional instances stay clear of the irruption threshold | Mean distance to θ_irr across archetypes: 0.42–0.61 (population mean, s forced to 0.05) |
| Functional archetype ranking is clinically coherent | Population-mean functional rigidity: Paranoia 0.644 > Hysteria 0.569 ≈ Obsession 0.568 > Melancholia 0.486 — paranoia remains the most rigid even without active crisis |
| Crisis instances (s=1) reproduce the original validated means | Population means at s=1 match the Section 3 symptomatic anchors to within 0.02 across all seven state variables |
| Domain and determinism | 0 violations across 10,000+ samples; identical seed reproduces identical state, including the sampled intensity |

The four named demonstration instances (Valentina, Martín, Elena, Roberto) remain fixed at s=1 to preserve the validated clinical vignettes of Section 7.1 unchanged; the intensity axis is exposed as a user-controlled parameter for new instances and for casual, non-crisis interaction — and, as detailed in Section 8.7, for population-level synthetic samples in the social influence layer.

## 7. Preliminary Evidence and Validation Protocol

### 7.1 Demonstration Sessions

We have run demonstration sessions for all seven archetypes under a student-of-psychology interaction protocol. Sessions consist of 4-6 exchanges with no explicit behavioral prompting — the interviewer asks open questions; the archetype responds according to its structural parameters. Key findings:

- Irruption events occur without scripting, as emergent consequences of $\Pi$ exceeding $\theta_{irr}$. The form of irruption is archetype-specific: hysteria closes behaviorally, obsession ritualizes, paranoia includes the interviewer.
- $\rho$ (fantasy rigidity) shows consistent decrease across sessions for neurotic archetypes — a computational analogue of therapeutic process. Paranoia shows minimal $\rho$ decay, consistent with clinical theory.
- $\sigma_{fan}$ activation correctly tracks theoretically relevant questions: the hysteria instance's fantasy object activates on choice/abandonment questions, the paranoia instance's on contingency challenges.

### 7.2 Blind Clinical Evaluation Protocol

We have designed and are executing a Level-1 validation study: blind classification of ANIMA transcripts by clinical evaluators with psychoanalytic training. Protocol summary:

| Parameter | Value |
|---|---|
| Corpus | 7 transcripts (one per archetype), 4 exchanges each, presented in randomized order without structure labels |
| Evaluators | N = 10-15; inclusion criteria: practicing psychoanalytic clinician or advanced trainee |
| Task | Classify structure (7 options), rate confidence (1-5), describe markers, indicate differential if confidence ≤ 3 |
| Primary measure | Classification accuracy vs. chance ($1/7 \approx 14.3\%$); binomial exact test, $p < 0.05$ |
| Secondary measure | Inter-rater kappa (target: $\kappa \geq 0.30$); confusion matrix for development guidance |
| Success threshold | Accuracy ≥ 40% (≈ 3× chance) AND $\kappa \geq 0.30$ |

A positive result provides publishable empirical evidence of structural distinguishability. A negative result provides a confusion matrix that precisely identifies which archetypes require further engine development — a result equally valuable for the research program.

## 8. Applications

### 8.1 Agent Behavioral Evaluation

ANIMA's DSE is a prototype methodology for auditable behavioral evaluation. Current agent evaluation frameworks assess surface properties (task completion, factual accuracy). Structural consistency over time — whether an agent's behavioral trajectory is coherent, whether it maintains appropriate internal pressure, whether its responses are generated from an actual internal state — is not evaluated. The DSE's deterministic signal extraction provides a framework for this evaluation that requires no additional LLM calls and produces traceable, reproducible results.

### 8.2 Structured Synthetic Data Generation

Training data for conversational agents is predominantly flat: behaviorally undifferentiated, without internal coherence, without the dynamics of accumulating pressure and release. ANIMA can generate structurally diverse conversational corpora — seven distinct behavioral organizations, each with internally consistent trajectories across sessions. Synthetic data generated through ANIMA carries behavioral structure that current synthetic data generation cannot provide.

### 8.3 Clinical and Training Applications

Clinical training in psychoanalytic or psychodynamic psychotherapy requires exposure to diverse structural presentations — not always available in supervised practice. ANIMA can generate training cases with calibrated structural properties, including controlled irruption events at known turn numbers, for supervised learning environments. The blind evaluation protocol we have designed functions simultaneously as a validation study and as a training assessment tool.

### 8.4 Persistent NPC and Character Systems

Game and interactive narrative systems require characters that behave consistently over long interactions and accumulate history. Current LLM-based NPCs lose behavioral coherence beyond a few turns. ANIMA's architecture provides a principled solution: persistent internal state that the LLM renders, rather than state maintained by the LLM itself.

### 8.5 Market Simulation and Behavioral Economics (Future Direction)

Current agent-based market simulation models face a structural limitation: their agents either follow simple behavioral rules or receive a one-time personality prompt from an LLM. Neither approach can model the accumulation of internal pressure across repeated exposures, the evolution of a subject's position over time, or the differential response to messages based on the agent's underlying subjective organization. The result is simulations that answer demographic questions rather than psychological ones.

ANIMA's psychodynamic engine is a natural candidate for richer simulation. Instead of asking 'what percentage will buy?', a market built on ANIMA's architecture could address structurally different questions:

- Which message generates initial adhesion across different subjective organizations — and which produces immediate resistance?
- Which profiles change position after repeated exposure, and which develop resistance after initial interest?
- How does acceptance evolve after multiple campaign interactions over time?
- Which segments become promoters, and what internal state profile predicts that transition?

In a streaming service launch scenario, for example, a synthetic market of subjects with diverse psychodynamic state distributions could model how different campaign messages interact with different internal economies — not as statistical correlations but as structural responses derived from the agent's state. A subject with high Π (unconscious pressure) and low ρ (fantasy rigidity) responds differently to a scarcity message than a subject with high G (jouissance) and high ρ. That difference is not a personality trait — it is a dynamic property of the agent's internal state at the moment of exposure.

This distinguishes the approach from current digital twin and consumer simulation platforms, which model behavior as a function of demographic attributes or static preference rules. ANIMA models behavior as a function of a persistent, evolving internal state — which means the same message, delivered to the same agent at different moments in a trajectory, can produce different outcomes.

**Current status:**

This application requires infrastructure development beyond the current prototype — specifically, a backend capable of running thousands of concurrent instances with inter-agent social influence modeling. We are currently focused on validating the core architecture in the clinical domain, which provides the empirical foundation for market simulation applications. We are open to exploring this direction with partners who have simulation infrastructure and interest in richer behavioral modeling.

### 8.6 Political Communication and Discourse Research (Future Direction)

Political communication research faces a methodological constraint: studying how different message types interact with different psychological profiles requires either large-scale human subject studies — expensive, slow, and ethically constrained — or simplified computational models that lack psychological depth. ANIMA offers a third option: a simulator of psychological responses to discourse, not a predictor of individual decisions.

The distinction is precise and consequential. ANIMA can model how a given type of subjective organization responds to a given type of message — which profiles show greater resistance, which show greater openness, which tend toward polarization. It cannot and should not be used to identify real individuals with those profiles and target them accordingly. The unit of analysis is the structural type, not the person.

Legitimate experimental uses under this framing include:

- Exposing the same political message to thousands of simulated subjective states and observing which profiles show greater resistance, greater adhesion, or greater polarization — as a tool for understanding structural vulnerabilities in public discourse, not for exploiting them.
- Comparing how different communication styles generate different internal responses by subjective organization — studying what makes a message land as persuasion versus imposition versus threat, as a function of the listener's internal state at the moment of exposure.
- Studying which types of discourse favor dialogue versus confrontation — identifying structural features of communication that increase or decrease the probability of genuine elaboration in the interlocutor.

These applications are closer to computational political psychology than to political strategy. The research question is structural: what does a given type of discourse do to a given type of internal organization? Not: how do we reach and persuade this specific population?

**Ethical boundaries and acceptable use**

The political communication application of ANIMA raises risks that must be named explicitly. We identify three uses that fall outside the acceptable scope of this architecture:

- Psychological microtargeting: using ANIMA-derived profiles to identify and target real individuals or populations based on inferred internal state. This application is excluded from any collaboration or licensing arrangement.
- Influence campaign design: using ANIMA simulations to optimize persuasion strategies for deployment against real populations. The architecture is designed for understanding structural responses, not for engineering them at scale.
- Predictive profiling: presenting ANIMA outputs as predictions of individual behavior or decision. The system models structural tendencies, not individual trajectories.

The distinction between a simulator of structural psychological responses and a tool for behavioral manipulation is architectural — it depends on whether the output is used to understand patterns or to target persons. We commit to this distinction as a condition of any collaboration involving this application domain, and we welcome institutional review for research uses in political communication.

**Worked example: structural response to public-health messaging**

As a bounded illustration of the acceptable-use distinction above, we ran a synthetic population (N=3,000, realistic base rates from Section 8.7) through three generic public-health message types — authoritative (“the health ministry recommends vaccination”), consensus-based (“89% of the population is already vaccinated”), and dialogic (“let's talk through your concerns”). No real campaign, institution, or population was modeled; the message texts are generic placeholders chosen for their discursive type, not their content.

| Archetype | Structural pattern |
|---|---|
| Paranoia | Weakest response to the authoritative message, strongest to the dialogic one — consistent with authority-as-threat in the archetype's social permeability profile (Section 8.7). |
| Obsession | Strongest response to the authoritative message — consistent with procedural respect for institutional authority. |
| Hysteria | Strongest response to the consensus message — consistent with high permeability to group and consensus signals. |
| Melancholia | Strongest response to the dialogic message — the only message type that makes no demand, consistent with low baseline energy. |

This is the intended scope of the application: a structural finding about which discourse types different internal organizations resist or accept — not a recommendation for which message to deploy against a given population. No individual-level targeting, no persuasion optimization, and no real-world message or actor was involved in producing this result.

## 8.7 Social Influence Layer (Prototype, Validated — v2)

A fourth layer extends the architecture from a single subject in dialogue to a population of subjects embedded in a social graph. This layer does not modify the psychodynamic engine described in Section 3 — it modulates the signal vector $\sigma(t)$ that feeds it. This preserves the core architectural principle: the engine remains a pure, auditable function; social context enters only as an additional signal contribution, computed by a separate deterministic module.

Each subject in the population carries the same seven-variable state $S(t)$ as in Section 3, plus a scalar position $p(t) \in [0,1]$ representing adherence to an external message (e.g., a campaign, a policy announcement, a product launch). At each simulation tick, five factors modulate how strongly a message shifts $p(t)$ for a given subject:

- Authority — weight of the message source's perceived authority.
- Consensus — perceived proportion of the population already adhering to the message.
- Group influence — mean position of the subject's direct neighbors in the social graph.
- Source trust — accumulated reliability of the specific message source for this subject.
- Reputational cost — social cost of diverging from the perceived majority position.

Critically, each archetype has a distinct permeability profile across these five factors — derived from the same structural logic that governs the engine's defense mechanisms. Hysteria, organized around recognition by the Other, shows high permeability to consensus and group influence. Obsession, organized around procedural control, responds to authority but is largely indifferent to consensus. Paranoia inverts the sign: authority and consensus actively repel rather than attract, consistent with a structure organized around the Other's malevolent intent. The same campaign, delivered with the same strength, produces structurally differentiated population-level outcomes — not because of randomness, but because of the receiving structure's characteristic relation to the social field.

### v2 refinements: realistic prevalence and individual permeability

An initial implementation assigned archetypes to population nodes uniformly (25% each), and gave every node of a given archetype identical social permeability. Both simplifications were removed in v2, using the same intensity-graded sampling machinery introduced in Section 6 (structural archetype × clinical intensity).

First, archetype assignment now follows approximate population base rates rather than a uniform split — directionally realistic rather than epidemiologically literal: common neurotic presentations (hysteria, obsession) are frequent; a fully clinical paranoid structure is rare. Second, each node's own sampled rigidity ($\rho$) and anxiety ($A$) — not just its archetype's constant profile — modulate its susceptibility: higher individual rigidity dampens group- and trust-based pushes; higher individual anxiety amplifies authority- and consensus-based pushes. Two nodes of the same archetype can therefore respond differently to the same message, exactly as the intensity axis intends.

### Validation (v2)

The v2 dynamics were re-validated across five tests, run deterministically with seeded random generation for reproducibility:

| Test | Result |
|---|---|
| Population composition (N=2,000) vs. target base rates | Hysteria 32.0% (target 32%), Obsession 33.3% (33%), Melancholia 24.4% (25%), Paranoia 10.3% (10%) — all within 0.6 percentage points |
| Individual variance within one archetype (Obsession) | $\rho$ (rigidity) range of 0.67 between the most rigid and most functional node of the same archetype — confirms real within-structure individuality, not archetype-constant behavior |
| Stability (36 nodes, 100 ticks, 3 campaign strengths) | Converges in all three conditions; late-window variance ≤ 0.00008 |
| No runaway polarization (max campaign strength) | 0 of 36 nodes reach extreme positions (<0.05 or >0.95) |

| | |
|---|---|
| Determinism | Identical seed reproduces identical population composition and trajectory, byte-for-byte |

A working prototype implements this layer over a 36-node small-world graph (ring lattice with random long-range rewiring, following Watts-Strogatz topology), with a subset of nodes designated as high-authority influencers. The prototype renders the propagation of a message in real time, with per-archetype and per-individual adherence tracked as the simulation runs.

**Relation to Section 8.5**

This layer is the mechanism that makes the market simulation application (Section 8.5) concrete rather than aspirational: it is what allows a population of psychodynamically distinct synthetic subjects to respond differentially, and propagate that response through a social structure, to a single piece of communication — which is the operational definition of the questions posed in that section (which message generates adhesion, which produces resistance, how acceptance evolves).

### 8.7.1 Additional Structural Experiments

Beyond the discourse-response worked examples above, we ran four further experiments on the v2 social layer, each isolating a different structural question. All reuse the validated engine and parameters unchanged — only simulation-level conditions (repetition, network wiring, message assignment, population composition) vary.

| Experiment | Finding |
|---|---|
| Repeated exposure (4 waves of the same message, N=200) | Adherence gain drops from +0.081 (wave 1) to ≈ 0 by wave 3. This is a mechanical property of the model's convergence dynamics — the population reaches a fixed point under constant message parameters — not emergent psychological fatigue, which would require an explicit decay term not currently implemented. |
| Cross-message exposure (split population, shared network, N=120) | Border nodes (adjacent to the other message group) shift measurably toward that group's mean relative to interior nodes of their own group, in both directions. A modest but directionally consistent boundary effect, confirming that peer influence propagates across message-exposure boundaries independent of each node's own direct message. |
| Population composition sensitivity (default vs. elevated paranoia prevalence, N=2,000) | Aggregate adherence to an authoritative message drops from 0.577 to 0.542 (default 10% vs. 30% paranoia share) with the message held fixed. Confirms population composition is a first-order driver of aggregate outcome, independent of message design. |

### A refined null result: network topology

An initial network-topology experiment (three rewiring probabilities, comparing final aggregate mean and a coarse convergence-tick estimate) found no measurable difference across topologies. Rather than treat this as evidence that topology is irrelevant, we treated it as underpowered: aggregate mean is, by construction in diffusion models generally, one of the metrics least sensitive to network structure. We re-ran the experiment with two changes: a between-node dispersion metric (standard deviation of adherence across the population) in place of the aggregate mean alone, and substantially higher topological contrast — a genuinely clustered structure (four dense communities of 50 nodes with only one bridge edge per community pair) against a near-random graph (Erdős–Rényi-style, average degree 4), with the small-world default from Section 8.7 as a middle condition.

| Topology | Final std. dev. across nodes |
| --- | --- |
| Clustered (4 communities, minimal bridging) | 0.0469 |
| Small-world (default) | 0.0470 |
| Near-random (average degree 4) | 0.0468 |

The refined experiment confirms the null result rather than overturning it: dispersion across topologies differs by 0.0002, and even the four communities in the clustered condition show near-identical internal means (0.570–0.592) — no echo-chamber effect emerges. We can now explain why, rather than merely report that it happens: in the current weighting, the authority and consensus terms reach every node identically as a broadcast signal and are averaged with the group term at equal weight; broadcast-style influence dominates the aggregate over peer-to-peer structure, making the network's wiring nearly irrelevant to outcomes at the population level. This is a legitimate, mechanistically explained property of the present parameterization — not evidence that network structure is unimportant in general. A model in which topology is intended to matter more would need to increase the relative weight of the group term, or model influencer reach as embedded within specific communities rather than uniformly accessible, both of which are direct, identifiable parameter changes rather than open research questions.

### 8.7.2 A Habituation Mechanism, Motivated by Experiment 1

Experiment 1 (repeated exposure) showed adherence saturating almost immediately — but as noted there, this was a mechanical consequence of convergence to a fixed point under a constant message, not modeled habituation. We implemented an explicit, stimulus-specific habituation mechanism to make genuine decay simulable and testable, distinct from mere convergence.

Each node carries a habituation level $H \in [0,1]$ and remembers the identity of the last message it received. If the current message matches the previous one, $H$ rises asymptotically toward 1; if the message changes, $H$ falls sharply (partial dishabituation via novelty) — consistent with the classical habituation/general-fatigue distinction (Groves & Thompson): habituation is stimulus-specific, not a generic exhaustion.

A first implementation applied H as an attenuation on the net push before computing the target position. This was a design error we caught in validation: attenuating the push shrinks the target itself, which pulled already-habituated nodes backward, away from progress they had already made — an artifact with no behavioral justification. The corrected design leaves the target computation untouched (always computed from the fresh, unattenuated push) and instead attenuates only the convergence rate toward that target. Habituation now slows approach; it never reverses it, and a bounded maximum attenuation (0.85) ensures no node becomes fully insensitized.

We then re-ran all four experiments from Section 8.7.1 with habituation enabled, holding every other parameter, seed, and topology identical to the published runs, to check whether the new mechanism overturns any prior conclusion.

| Experiment | Effect of enabling habituation |
|---|---|
| 1 — Repeated exposure | Wave-by-wave gains no longer collapse to ≈ 0: [0.0652, 0.0080, 0.0040, 0.0021] vs. the original [0.0808, 0.0008, 0.0000, 0.0000]. The population keeps crawling toward the target instead of freezing. Conclusion refined, not reversed. |
| 2 — Topology (refined) | Dispersion across topologies remains flat (0.0448 / 0.0449 / 0.0447 vs. the original 0.0469 / 0.0470 / 0.0468) — slightly lower overall (less time to converge within the fixed tick budget) but the topology-insensitivity finding is unchanged. |
| 3 — Cross-message exposure | The boundary effect persists in direction and comparable magnitude (Group A: interior 0.578 vs. border 0.564; Group B: interior 0.521 vs. border 0.525) against the original (0.587/0.571 and 0.524/0.529). Conclusion confirmed. |
| 4 — Population composition | The elevated-paranoia gap persists (3.0 percentage points vs. the original 3.5) — smaller because habituation slows both conditions similarly within a fixed tick budget, not because the composition effect weakened structurally. Conclusion confirmed. |

No prior finding is overturned by adding habituation. Its distinctive signature is specifically visible in Experiment 1's wave-by-wave pattern — exactly the phenomenon it was built to capture — while acting as a modest, uniform slow-down elsewhere, since Experiments 2–4 were designed around converged end-states rather than the transient, message-repetition dynamics habituation targets.

**Worked example: commercial persuasion mechanisms**

As a second bounded illustration, we ran the same synthetic population (N=3,000, realistic base rates) against three generic commercial persuasion mechanisms for a hypothetical product launch — scarcity (“only a few spots left”), belonging (“join the community that already has it”), and exclusivity (“access reserved for a few”). No brand, company, or real campaign was modeled; each mechanism was mapped to the psychodynamic variable it most plausibly recruits: scarcity to anxiety (A), belonging to libidinal bond (E), exclusivity to jouissance (G).

| Archetype | Structural pattern |
| --- | --- |
| Hysteria | Strongest response to belonging — consistent with organization around recognition by the Other. |
| Obsession | Second-strongest response to belonging, weakest to scarcity — procedural control dampens urgency-driven appeals more than social ones. |
| Melancholia | Strongest response to belonging, weakest to exclusivity — a structure organized around diminished self-worth shows little pull toward being “special.” |
| Paranoia | Strongest response to exclusivity, clearly above the other three archetypes — distinction from the majority is structurally coherent with a subject organized around not being like others. |

Belonging and exclusivity produced clear structural differentiation (a 0.08–0.09 range in population-mean response across archetypes). Scarcity produced a substantially smaller range (≈0.03) across two independent parameterizations of the anxiety-urgency link. We report this as a finding rather than adjusting the model until it conforms to an expected pattern: scarcity appeals may recruit a comparatively universal alarm response that is less mediated by structural position than appeals organized around the subject's relation to the Other (belonging, distinction). Whether this holds beyond the current engine calibration is an open question for the validation work in Section 7.2, not a claim we consider settled.

## 9. Current Status and Development Roadmap

| Phase | Status |
| --- | --- |
| Psychodynamic engine (7 variables, update equations, irruption mechanism) | Complete — implemented in JavaScript, running in browser |
| LLM coupling protocol (Instruction Object, Pattern Library, system prompt builder) | Complete — exercised across three provider environments (Section 4.1): Claude Sonnet 4.6, Claude Sonnet 4.5, Gemini 2.0 Flash |
| Deterministic Signal Extractor v1 | Complete — structural regex + keyword matching + cosine similarity, zero LLM calls |

| | |
|---|---|
| Behavioral archetypes (7) | Complete — parameter bundles with cultural world calibration for Argentine urban context |
| Blind clinical evaluation — corpus | Complete — 7 transcripts, randomized, formatted for evaluators |
| Blind clinical evaluation — execution | In progress — evaluator recruitment via clinical collaborator network |
| Inter-session memory P(n) and structural plasticity | Specified, not yet implemented |
| DSE v2 — lightweight embeddings for semantic proximity | Specified, not yet implemented |
| Social influence layer (5-factor modulation, 36-node graph prototype) | Validated — stability, differentiation, no-runaway, determinism all confirmed |
| Multi-instance interaction (transference as co-construction) | Research stage |

The current implementation runs as a single-file React application with API calls to Claude claude-sonnet-4-6 as the renderer. The engine, DSE, and coupling protocol are implemented in pure JavaScript with no external dependencies beyond the React framework. The system is functional and demonstrable today.

## 10. Collaboration Opportunities

We are seeking research collaboration or sponsored development engagement with AI labs interested in the following problem spaces:

- Behavioral consistency evaluation: applying DSE-derived methodology to evaluate structural coherence in existing agent systems.
- Structured synthetic data: using ANIMA to generate behaviorally differentiated conversational corpora for agent training.
- Architecture research: extending the decoupling principle — deterministic state engine + LLM renderer — to other behavioral domains.

We are open to research collaboration agreements (with IP retained by project), sponsored research engagements, and technical consulting arrangements. We are not seeking acquisition at this stage.

**Contact:**

Project ANIMA — available for technical exchange and demonstration. Interested parties may request access to the full technical specification, live demonstration of the running system, and the validation protocol materials.

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